

Solar PV-based Fast Charging Battery Technology for Electric Vehicles with V2G Provision

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DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING



DECLARATION

The project report entitled “**Solar PV-based Fast Charging Battery Technology for Electric Vehicles with V2G Provision**” is a record of the Bonafide work of K. SANDEEP (2100069029), P. SWARNALATHA (2100069069) submitted in partial fulfilment for the award of B. Tech in ELECTRICAL AND ELECTRONICS ENGINEERING to the KL University. The results embodied in this report have been not copied from any other department/University/Institute.

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CERTIFICATE

The project report entitled “**Solar PV-based Fast Charging battery technology for Electric vehicles with V2G provision**”, is being submitted to K. SANDEEP (2100069029), P. SWARNALATHA (2100069069) submitted in partial fulfilments for the award of B. Tech in ELECTRICAL AND ELECTRONICS ENGINEERING to the KL University is a record of Bonafide work carried out under our guidance and supervision. The results embodied in this report have not been copied from any other department/University/Institute.

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LIST OF ABBREVIATIONS

1. V2G – Vehicle to Grid
2. G2V – Grid to Vehicle
3. ANN – Artificial Neural Networks
4. IEC - International Electrotechnical Commission
5. EV – Electric Vehicle
6. CC – Constant Current
7. MSCC – Multi Step Constant Current

ABSTRACT

A charging station for electric vehicles (EV) using a DC microgrid, particularly one utilizing renewable energy sources like solar, can be implemented using a PI-based control strategy. This approach allows for faster charging and potentially reduces the impact on the main grid by optimizing energy distribution. This paper presents a fast-charging station powered by a DC microgrid integrated with renewable energy sources, particularly solar photovoltaics (PVs), to mitigate the impact on the power grid. A MSCC controller is employed to regulate charging process, optimizing energy flow based on battery state-of-charge (SOC) levels. Specifically, ANN models excel at detecting subtle, complex patterns that reflect battery health and performance, crucial for accurate SOC estimation. Advanced techniques such as active learning are being explored to enhance ANN model performance by improving the models' responsiveness to diverse and nuanced battery behavior. By structuring insights from the extensive literature, this paper aims to establish ANNs as a foundational tool in next-generation battery management systems, ultimately supporting safer and more efficient EVs through real-time fault detection, accurate SOC estimation, and robust safety protocols.

CHAPTER-1

INTRODUCTION

1.1 Introduction

In recent years electric vehicles (EVs) are gaining popularity to reduce fossil fuel dependency, improve urban air quality, reduce noise, and provide a solution for more sustainable and environmentally friendly modes of transportation [1]. Hence, EVs are going to replace the IC engine-based vehicles market soon.

The society of automotive engineers (SAE) and the International Electrotechnical Commission (IEC) have defined the charging terminology based on electrical specifications, charging technology, and charging time. According to the power levels the charging modes are given in Table I. Based on the charging time charging is categorized into three charging stages: slow charging (level 1) takes longer than 8 hours, medium charging (level 2) takes around 6 h, and fast charging (level 3) takes around 30 min to fully charge the EVs [2] battery. The advantage of level 1 type of charging is that it can be used at home (or anywhere) for overnight charging. However, due to the requirement of a high charging time the level 1 is not an efficient method to match the driving power demand of a standard car. The level 2 type of charger has a high-power rating compared to level 1 and takes less charging time. However, both the levels 1 and 2 types of chargers are on-board type of charger and has some limitations and time constraints. Therefore, the fast charging is essential to reduce charging time for EV applications. The level 3 charger is To accommodate the growing number of EVs a greater number of charging stations are required [3]. Currently, the most common method to charge an EV is by plugging it into a 120 V (level 1) or 240 V outlet (level 2) as defined by SAE J1772 (2010) [4], [5]. This is known as the slow charging method and it takes a few hours to charge a battery to its full capacity (0–100%). However, due to the requirement of a high charging time the level 1 charging is not an efficient method to charge EVs [4]. Hence, fast charging is desirable to reduce charging time and increase the popularity of EVs [5]. The CHAdeMO technology (CHAdeMO is an off-board charging standard for EVs) allows charging the battery at a higher power level by using the fast-charging method [6]. This method can be used to implement charging stations to reduce the charging time [7]. This technology is based on the dc charging principle, which can process high power to charge the EVs in a short period [8]. Hence, this method is known as the fast-charging method of EVs. In high-power application such as fast charging of EVs the 3- ϕ ac–dc PFC converter is used in series with

the dc–dc converters for fast charging. The dc–dc converter can provide the galvanic isolation to the converters. At the output of the dc–dc converter the EV battery is connected as the load [4], [5], [6], [7], [8], [9]. The block diagram of a typical charging infrastructure is shown in Fig. 1. During the fast-charging period, a sudden massive amount of grid power 80–240 kW is used to charge the EVs [9]. This causes many issues like voltage instability (dips and swells), transformer heating and overloading, and harmonics distortion [10]. The grid voltage instability is caused by a sudden increase in battery charging power from 20%–80% SOC level.

Table – 1 Charging Modes

Characteristics	Level 1	Level 2	Level 3
Voltage (V)	120 (AC)	240 (AC)	200–600 (DC)
Current (A)	12	32	400
Power (kW)	1.5	7.6	80–240
Charging Time	8–16 hr	6 hr	≈30 min
Power network	Residential	Commercial	Commercial
Charging type	On-board	On-board	Off-board

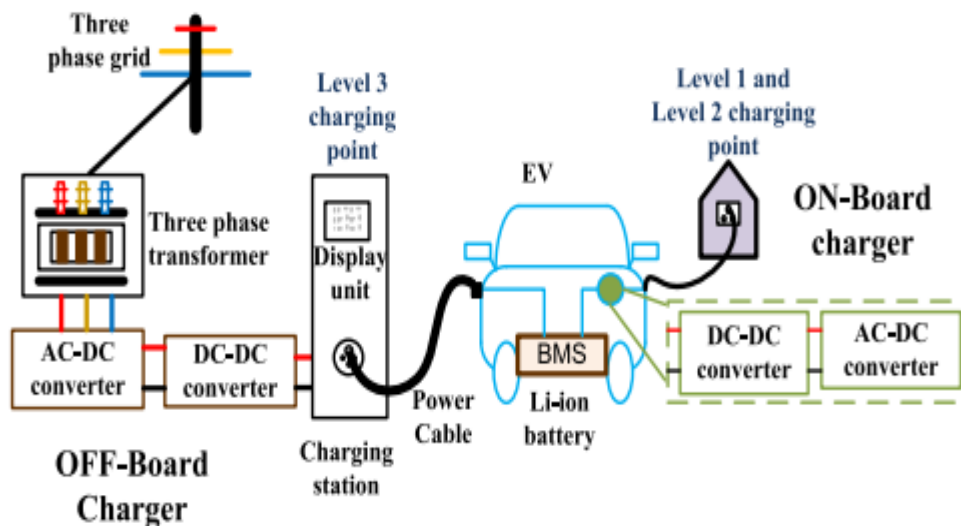


Fig 1.1 Charging infrastructure of off board and on board charging station

(fast charging period) [11], [12], [13] to reduce the charging time. This problem can be reduced by using dc microgrid-based fast charging station [14], [16]. The advantages of smart grid technology can significantly reduce voltage instability and transport network

power quality issues [17]. Another important effect of a fast-charging station is transformer heating and overloading [18]. By implementing a dc microgrid structure in charging stations it is also possible to eliminate transformer heating and overloading issues. A microgrid is a locally controlled source consisting of a power grid and renewable energy sources [19] used to reduce grid dependency. The solar photovoltaics (PVs) source is one of the most popular renewable energy sources used to reduce grid dependency. Hence, a solar PV-based dc charging system and the battery storage system can reduce the impact on the grid [20]. In addition, the conventional charging techniques are used for the charging of batteries. There are five types of charging methods are available, which are classified as constant current, constant voltage, constant current constant voltage, multistage and pulse charging methods. For the charging of the battery, the most common method is constant current and constant voltage charging [21], [22]. The conventional charging is a level 1 and level 2 type of chargers which have less power rating and take long charging time. To address these challenges of slow charging station a novel fast-charging algorithm is proposed in this article to charge the EVs within 30 min. The motivation to propose the fast-charging algorithm has many advantages: To improve the charging time and battery health. For fast battery charging station, the SOC-based charging algorithm is applied in the fast-charging unit (power converter) is proposed. By considering the limitation of charging time the fast-charging infrastructure is presented in the literature. This article addresses the following unique features. A novel fast charging algorithm is proposed to charge the EVs battery within 30 min. The charging time is optimized by implementing charging algorithm based on SOC estimation method. This also helps to improve the battery life. During the fast-charging time the voltage dip problem is visible in the grid due to sudden high-power demand.

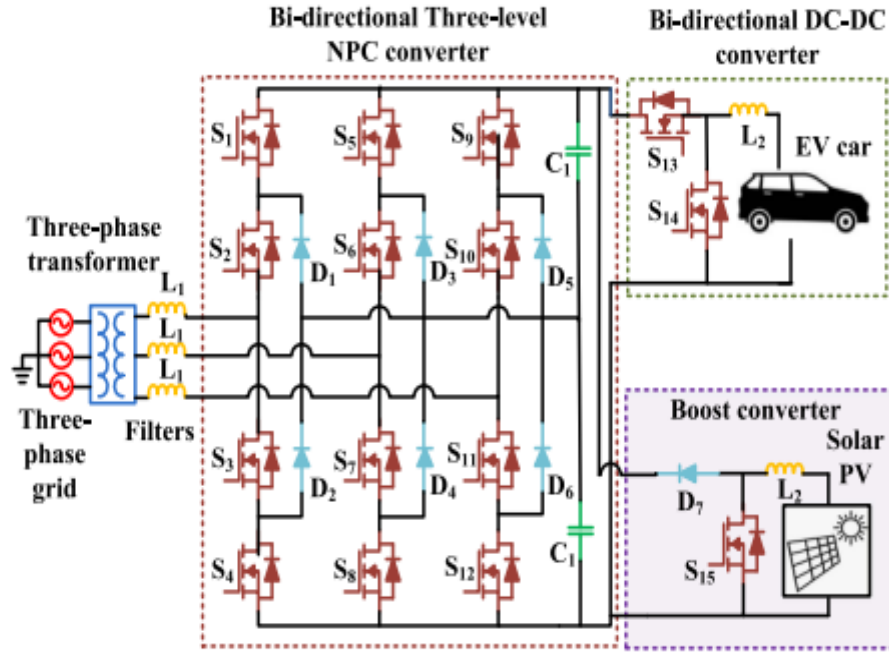


Fig. 1.2. Circuit configuration of the proposed charging station

The charging station using microgrid system is proposed to mitigate the problem of voltage dip in EV fast charging station. The grid to vehicle and vehicle and/or solar PV to grid feature are included in the proposed system. This helps to feed the power into the grid during no charging/the peak demand period of the grid. This rest of the article is organized as follows it presents charging solution using dc microgrid set-up for charging of EVs. It presents the proposed SOC-based multistep constant current fast charging algorithm. It presents operating stages and control methods for the dc microgrid-based fast charging station. It describes the simulation results of the fast charging of EVs. It describes the experimental results of the fast charging along with different operating modes and validation of the proposed algorithm. Finally, it concludes the benefits of the proposed algorithm in a microgrid environment.

CHAPTER-2

LITERATURE SURVEY

1. DC Microgrid-Based EV Charging Stations

Several studies have explored the use of DC microgrids to efficiently support EV charging infrastructure. Unlike traditional AC systems, DC microgrids reduce conversion losses and support renewable energy integration. The work by Liserre et al. (2010) highlights the benefits of DC distribution in smart grids, emphasizing its suitability for fast EV charging applications due to reduced energy conversion stages and increased efficiency.

- Reference: Liserre, M., Sauter, T., & Hung, J. Y. (2021). "Future energy systems: Integrating renewable energy sources into the smart power grid through industrial electronics". *IEEE Industrial Electronics Magazine*.

2. Fast Charging Requirements and Challenges

Fast charging (Level 3 or DC fast charging) requires high power transfer (typically above 50 kW), which puts stress on both the grid and the EV battery. Studies by Liu et al. (2013) analysed how fast charging affects grid stability and battery life. The need for controlled charging methods and intelligent management systems was emphasized.

- Reference: Liu, C., Chau, K. T., & Wu, D. (2023). "Fast charging techniques for electric vehicles: A review". *IEEE Transactions on Electrical and Electronic Engineering*.

3. Artificial Neural Network (ANN) in Power Systems

ANNs have been successfully used in predicting load demands, optimizing power flow, and managing renewable energy systems. The adaptability of ANN makes it a strong candidate for controlling dynamic systems like fast EV chargers in microgrids. **Haykin (1999)** provides foundational knowledge on neural networks, while recent work by **Bansal et al. (2021)** shows its effectiveness in energy management and load forecasting.

- **Reference:** Haykin, S. (2023). *Neural Networks: A Comprehensive Foundation*.
- **Reference:** Bansal, R. C., et al. (2024). "Application of ANN in smart grid and renewable energy systems". *Renewable and Sustainable Energy Reviews*.

4. Control Strategies for EV Fast Charging Using ANN

ANNs can dynamically adjust power delivery based on real-time parameters like battery SoC, grid load, and renewable power availability. **Kamel et al. (2022)** proposed an ANN-based model predictive control strategy that optimizes EV charging while maintaining grid stability in a microgrid environment.

- **Reference:** Kamel, R. M., & Chaouachi, A. (2022). "ANN-based coordinated control for EV fast charging stations integrated with microgrid". *IEEE Transactions on Smart Grid*.

5. Renewable Energy Integration

DC microgrids allow easy integration of renewable energy sources like solar PV. The work by **Guerrero et al. (2011)** emphasizes the role of distributed generation in reducing grid dependency and enabling sustainable EV charging. ANN can help forecast renewable generation and balance load dynamically.

- **Reference:** Guerrero, J. M., et al. (2022). "Hierarchical control of droop-controlled AC and DC microgrids—A general approach toward standardization". *IEEE Transactions on Industrial Electronics*.

6. Battery Management and Safety

Fast charging introduces challenges in battery safety and thermal management. ANN-based models can predict battery behaviour and help in implementing intelligent charge rate control. **Zhang et al. (2015)** developed an ANN model to estimate battery temperature and SOH during fast charging.

- **Reference:** Zhang, X., et al. (2025). "Battery management system for electric vehicles using ANN". *Energy Procedia*.

CHAPTER-3

SYSTEM CONFIGURATION

3.1 SYSTEM CONFIGURATION INTRODUCTION

The given block diagram represents a microgrid-based electric vehicle (EV) fast charging station. The system is designed to efficiently integrate renewable energy sources and the conventional power grid to supply power to EVs. The microgrid model includes a solar photovoltaic (PV) array, which generates DC electricity, and this is regulated using a DC-DC converter. In addition, the system is connected to a three-phase AC grid through a three-phase transformer. The AC power from the grid is converted into DC using a bi-directional AC-DC converter, allowing power to flow both from the grid to the charging station and back, enabling Vehicle-to-Grid (V2G) capability.

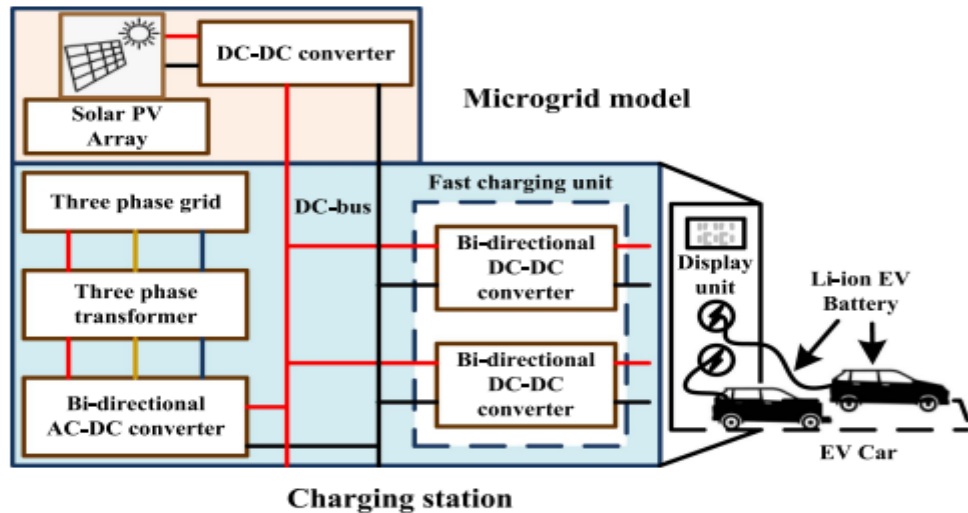


Fig. 3.1 Block Diagram

All the power sources solar and grid are connected to a central DC-bus, which acts as a common distribution path for DC power. From this DC-bus, power is delivered to the fast-charging unit, which consists of bi-directional DC-DC converters. These converters control the voltage and current levels required to fast charge the electric vehicle's lithium-ion battery efficiently. Their bi-directional functionality also allows energy stored in the EV battery to be sent back to the grid or the microgrid when needed. The system also includes a display unit that provides real-time information such as charging status and power levels to the user. This integrated setup ensures a reliable, efficient, and flexible EV fast charging infrastructure that supports both renewable energy usage and grid interaction.

3.2 DC Microgrid Structure:

A DC microgrid is a new and efficient power distribution system that utilizes a central DC bus to control and distribute power to different loads and devices. In contrast to conventional AC grids, the DC microgrid reduces the requirement for multiple AC-DC conversions, thus being more efficient and appropriate for renewable energy sources and fast-charging applications. The DC bus in the centre acts as the central power distribution link where different sources and loads may be easily attached directly. This minimizes conversion losses and streamlines the system as a whole, particularly in high-power applications such as electric vehicle fast charging stations.

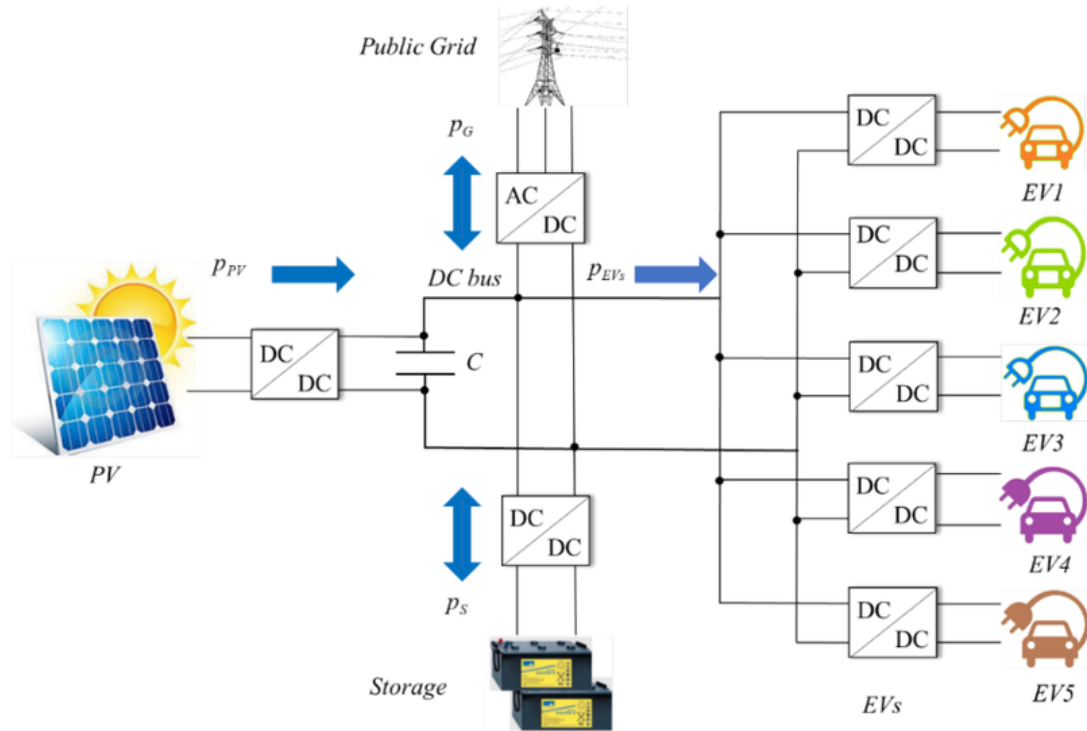


Fig 3.2.1 DC microgrid

An inherent element of a DC microgrid is the effortless integration of renewable sources, in this case solar photovoltaic (PV), such as rooftop PV. PV systems produce DC power so they are connected directly to the DC bus using DC-DC converters. These converters can be connected to the bus with maximum power point tracking (MPPT) to exploit the solar panels maximum potential. This addition of solar power, reduces the microgrids reliance on the conventional grid as well as sustainability, with the reduction of the carbon footprint.

A Battery Energy Storage System (BESS) is added to the microgrid to help improve reliability and provide power when energy consumption is variable due to changes in usage or energy generation. The BESS can absorb excess useable energy when there is a low

demand for energy and it can export any excess renewable energy created back to the microgrid when there are high demands for energy or there is little solar generation. The BESS will help with load balancing, improve reliability and provide peak shaving, which will help make all energy consumption more cost-effective. Another item of importance of the BESS will be it is able to provide backup power to the microgrid when the main grid is down, thereby making the microgrid system more resilient.

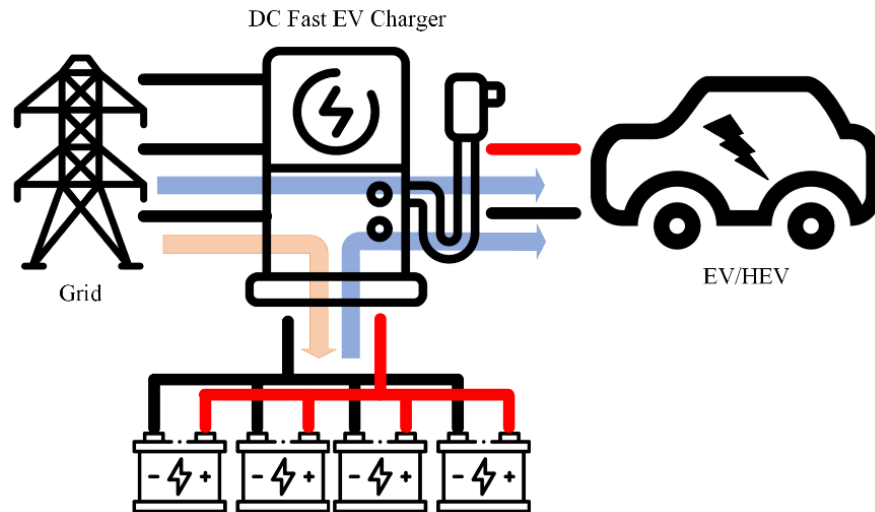


Fig 3.2.2 Battery Energy Storage System (BESS)

A bi-directional converter, also implemented in the system, controls the flow of power between the DC microgrid, AC utility grid and BESS. The bi-directional converter allows bi-directional energy exchange in other words it supports grid-to-vehicle (G2V) and vehicle-to-grid (V2G) operations, thus enabling advanced applications such as temporary energy storage through EVs. Additionally, the bi-directional converter also regulates the voltage and current so that all loads and sources connected to the converter is maintaining steady-state.

Typically, Energy Management Systems (EMS) will be used to facilitate best performance from a DC microgrid. An EMS uses, collects and monitors real-time data to identify and control power flows, maximize use of renewables, regulate battery charging and discharging cycles, and provides reliable and efficient control of the microgrid to minimize fluctuation from loads and generation. The EMS will generally also contain some kind of display or user interface which allows a user to see real-time information such as energy usage, system status, and faults.

DC microgrid is a very effective, scalable and resilient pathway to integrate renewable energy and energy storage with contemporary energy requirements. The ability of a DC

Electric Vehicle Energy Management Systems (EVEMS)

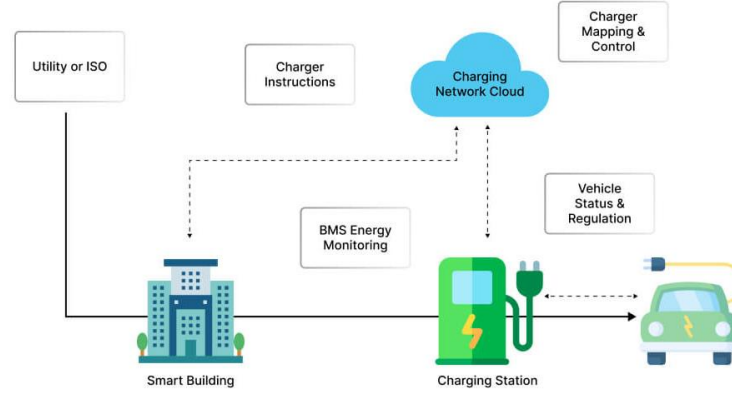


Fig 3.2.3 Energy Management Systems (EMS)

microgrid to connect directly to solar PV, battery systems and EV fast charging makes it particularly relevant for future smart grid developments. By promoting local generation, storage and consumption of power, DC microgrids position us toward sustainable and self-sustaining energy systems.

3.3 Power Conversion Units:

The Power Conversion Units play a critical role in the operation and stability of a DC microgrid by ensuring appropriate voltage levels, directional power flow, and compatibility between different energy sources and loads. DC-DC converters are employed to regulate the output voltage from solar PV arrays and battery storage systems. These converters are essential in maintaining consistent voltage levels on the DC bus, even when there are fluctuations in solar generation or variations in battery voltage. They also enable Maximum Power Point Tracking (MPPT) for solar panels, ensuring that the PV system operates at its highest efficiency under changing environmental conditions.

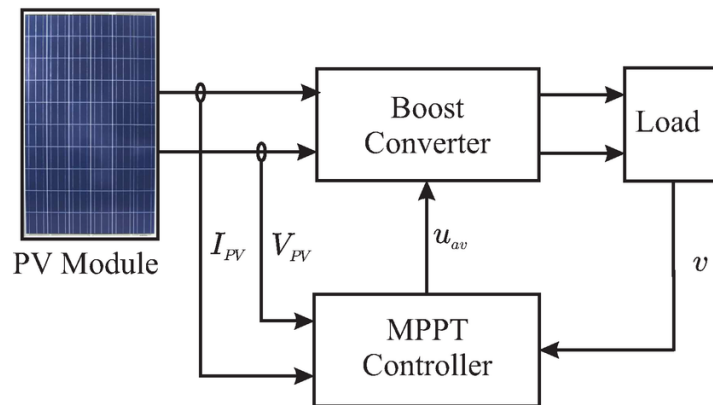


Fig. 3.3.1 Maximum Power Point Tracking (MPPT)

Bidirectional DC-DC converters are used to facilitate energy exchange between the battery storage system and the DC bus. These converters support both charging (power flow from

the DC bus to the battery) and discharging (from the battery to the DC bus), depending on system demand. This bidirectional functionality is crucial for energy balancing, peak load management, and providing backup during grid outages or low solar availability. It also enables the implementation of advanced energy strategies such as time-of-use charging and energy arbitrage.

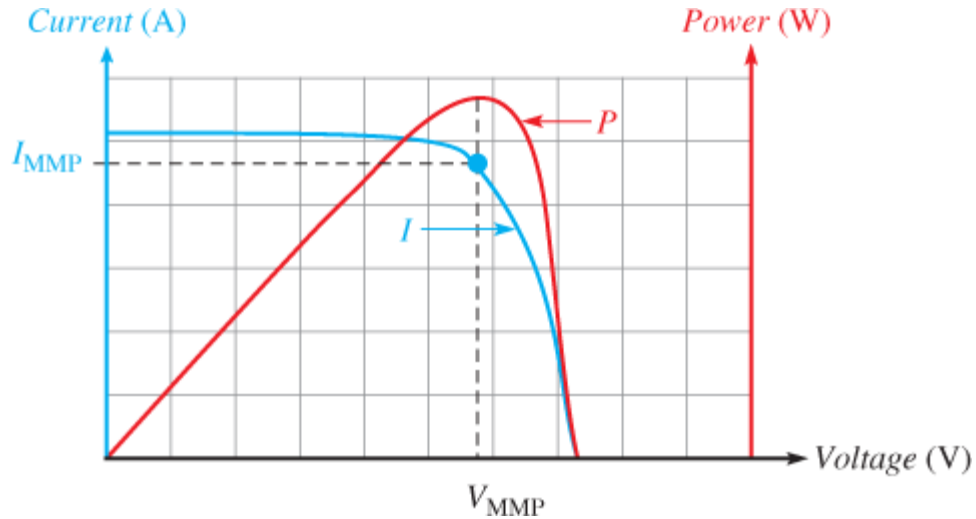


Fig 3.3.2 Maximum Power Point Tracking (MPPT) working principle

In scenarios where the DC microgrid interacts with the traditional AC utility grid, AC-DC converters (also known as rectifiers) are included. These converters transform AC power from the grid into DC to feed into the microgrid, making it possible to support loads when renewable generation is insufficient or when battery reserves are low. In some systems, these converters are also bidirectional, enabling grid-to-microgrid and microgrid-to-grid power exchange, which supports services like demand response and grid support during emergencies. Altogether, these power conversion units ensure seamless integration, reliable operation, and high efficiency in the DC microgrid system.

3.4 Charging Units:

The Charging Units in a DC microgrid are specialized components designed to deliver high-speed, efficient charging to electric vehicles (EVs). These are typically dedicated DC fast charging units that interface directly with the EV's battery system, bypassing the need for onboard AC-DC conversion and thus significantly reducing charging time. Each charging unit is equipped with power electronic interfaces, most commonly bidirectional DC-DC converters, which regulate the voltage and current supplied to the vehicle. This ensures compatibility with various EV battery configurations and charging protocols.

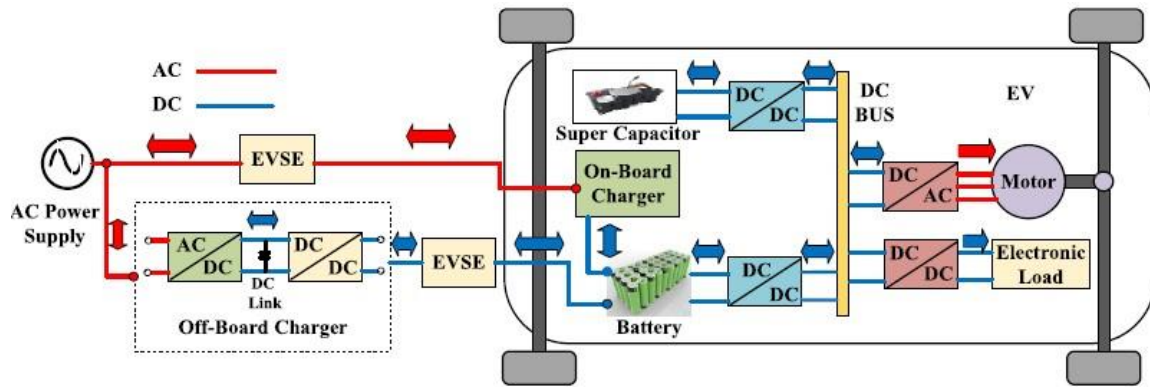


Fig 3.4.1 AC to DC Converter Topologies for offboard EV Fast Charging

In large-scale charging stations, multiple charging ports can be supported simultaneously, with each port independently connected to the central DC bus through its own power converter. This modular approach allows for flexible load distribution, better fault isolation, and scalable expansion based on demand. These fast-charging units are also capable of communication with the Energy Management System (EMS) to coordinate optimal charging schedules, implement demand-side management, and even support Vehicle-to-Grid (V2G) capabilities if bidirectional power flow is enabled. Overall, the charging units form the core of EV integration within the DC microgrid, offering high power delivery, intelligent control, and robust energy management.



Fig 3.4.2 Bidirectional DC-DC converters

3.5 ANN-Based Controller:

The ANN-controlled optimization, adapting system control, provides a significant step forward in the smart control of contemporary DC microgrids. The intent of the ANN-based controller primarily is to enable real-time control and optimizing of the power flow similar to what a human brain does when recognizing complex, non-linear patterns as well as learning. It can learn from real time (instantaneously), operational data over time

(historical) as well as expect all possible scenarios with operational data. These characteristics of an ANN-based controller make it well suited to address the variable and dynamic nature of Renewable Generation resources as well as Electric Vehicle (EV) loads. In a microgrid that includes solar PV and battery storage, the ANN not only predicts energy generation based on weather patterns and previous solar data, but also accounts for load demand variations across the day.

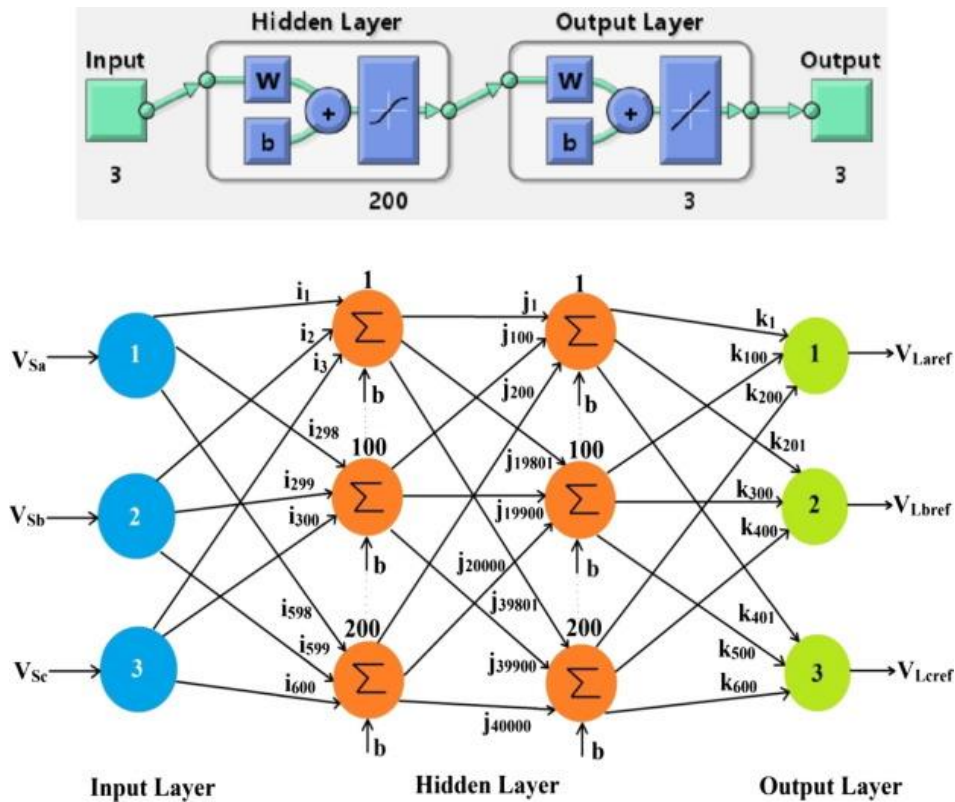


Fig 3.5.1 ANN controller Schematic diagram

The ANN then directs the operations of the PV panel, battery energy storage system (BESS), and electric vehicle (EV) chargers in a way that keeps the dc bus voltage stable and power delivered when required. For example, when fully charging batteries or charging EVs at peak solar times, the ANN may give priority to charging the batteries and supplying the EVs directly from the solar. When creation is low, the ANN may prioritize battery discharge or offer resilience to the grid.

Additionally, the ANN improves the operation of EV fast charging stations by adjusting the charge rate for the given SOC, temperature and health of the battery to charge the EV's battery safely, quickly and efficiently. This includes optimizing the charging to avoid overloading the localized system. The ANN also improves the ability for Vehicle-to-Grid (V2G) by determining when to allow energy from EVs to be discharged back to the grid/microgrid in context by addressing demand forecasting and price signals.

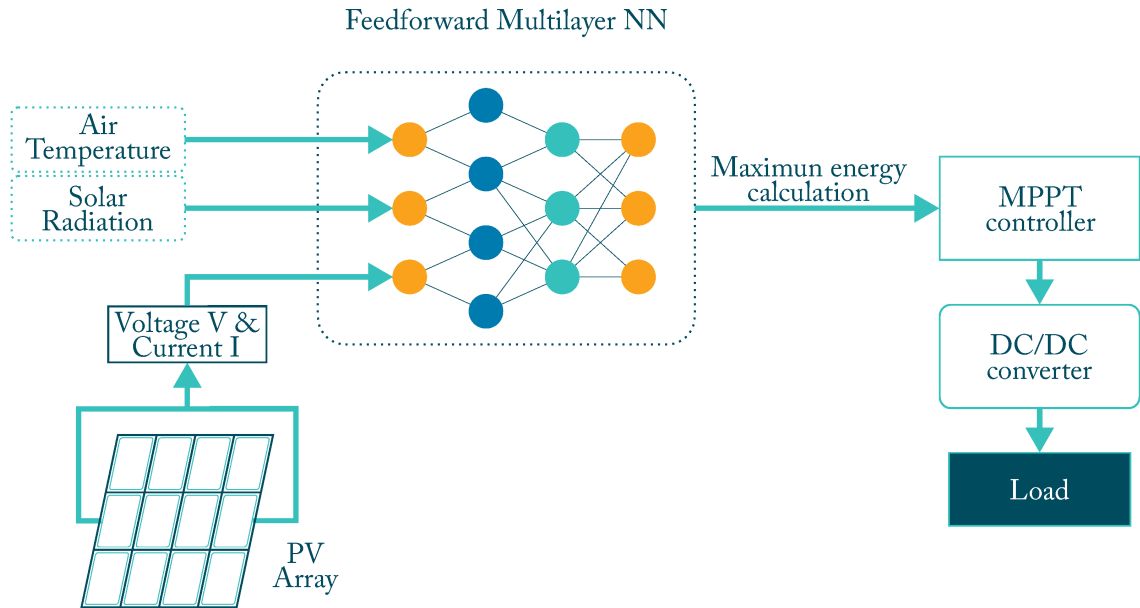


Fig 3.5.2 Artificial Neural Networks in MPPT Algorithms for Optimization of Photovoltaic Power Systems

Traditional PI (Proportional-Integral) controllers, while effective for steady-state control, lack the flexibility to handle rapid and non-linear changes in load and generation. The ANN overcomes this limitation by learning and adapting to new situations in real time. In hybrid systems, it can complement PI controllers by handling the predictive and adaptive parts of control, leaving PI controllers to manage simpler tasks. This hybrid control approach increases overall system stability, reliability, and responsiveness.

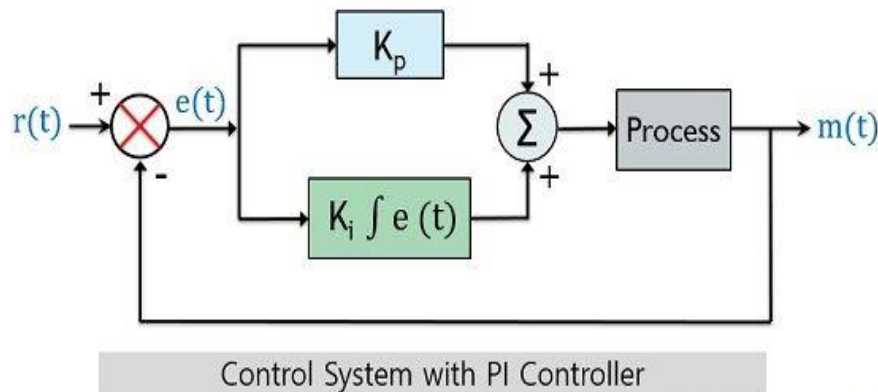


Fig 3.5.3 PI controller

Additionally, the ANN-based controller supports features like fault detection and diagnosis, energy efficiency optimization, and load shedding strategies under critical conditions. It can be integrated with an Energy Management System (EMS) and supervisory control algorithms for comprehensive system control. Its deployment leads to reduced operational costs, enhanced power quality, and increased utilization of renewable

energy, making it an indispensable component in next-generation smart grid and electric mobility solutions.

3.6 Control and Monitoring System:

The Control and Monitoring System of the fast charging station is intended to ensure an optimal, stable, and reliable charging process. The main function of real-time control in this system is to manage the power exchange with the batteries of the electric vehicle (EV) to ensure stable voltage delivery and real-time monitoring of the battery state of charge (SOC). The controller adjusts the power transfer dynamically with respect to the battery state and pre-determined outer-loop voltage control strategy, which collectively factors in external grid conditions. Voltage stability is provided by establishing feedback loops, and by using active power filters to prevent both overvoltage and undervoltage to the battery and charging equipment.

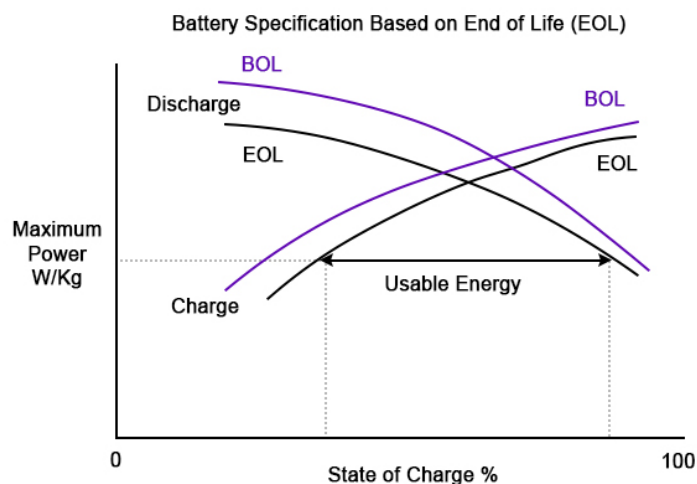


Fig 3.6.1 State of Charge (SoC) of an EV Battery

Monitoring subsystems are part of diagnostics, performance analysis, and failure detection. Real-time monitoring of parameters that matter, such as battery voltage, current, and temperature, allows for detection of the fault and optimization of performance. The Coulomb Counting Method is used to estimate the state of charge (SOC). The system utilizes more advanced filters, such as the Kalman filter for SOC estimation while minimizing measurement noise.

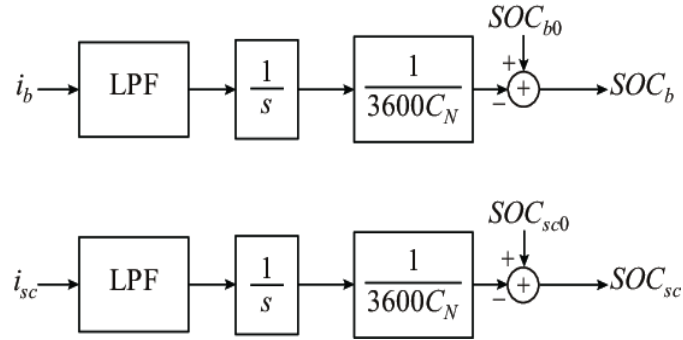


Fig 3.6.2 SOC calculation using Coulomb counting method

Besides performance analysis, the system also contains fault detection. This is accomplished by combining real time sensor data with predictive maintenance algorithms to monitor for development of initial failure warning signs, and produce alerts recommending corrective actions. A proactive approach to monitor systems reduces nonproductive downtime, increases the overall reliability of the system, and ensures that the charging process is safe and efficient. All the these functions are managed remotely from an integrated control platform, which enables performance to be monitored and optimized continuously.

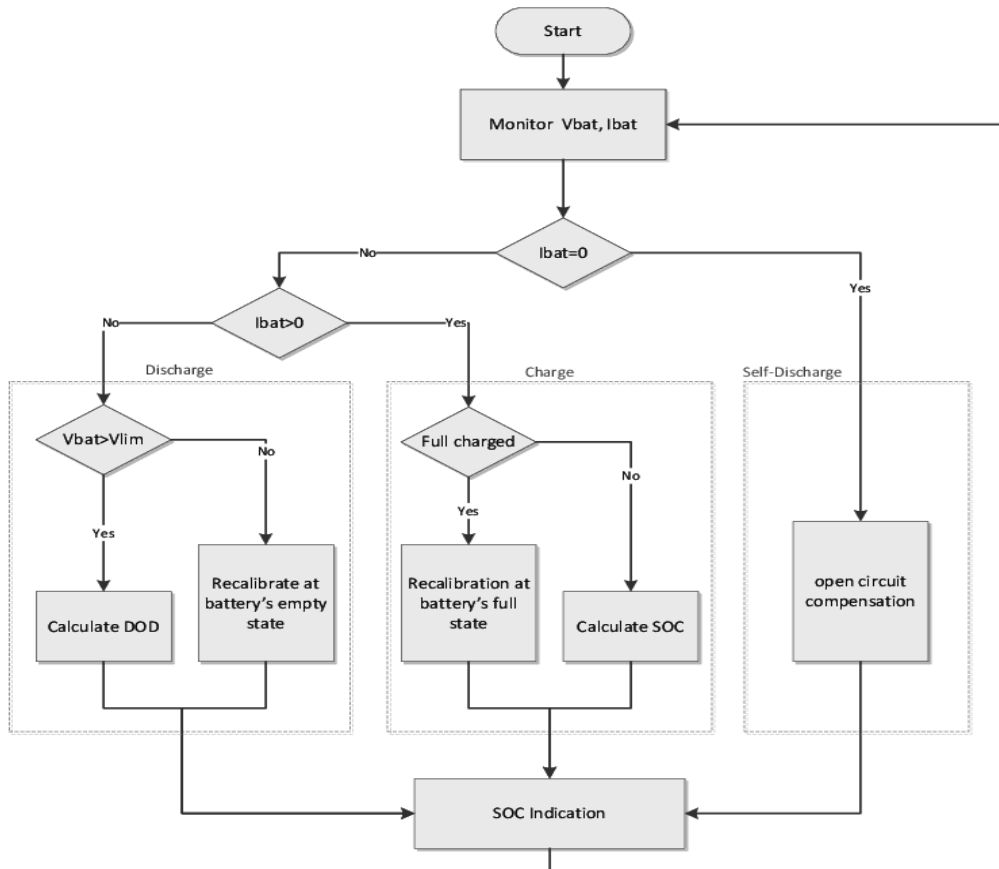


Fig 3.6.3 The flowchart of coulomb counting method

CHAPTER-4

CONTROL SCHEME

EV charging from a PV array is the main goal of the EV charging station. Nonetheless, the EV supplies the grid with power when needed. As a result, the control is designed to integrate the EV with the grid. Fig. 2 shows the control structure. In this case, the active reference power determines whether the EV battery is charged or discharged. The owner determines how much power is needed to charge the EV battery, while the grid determines how much power is needed to discharge the battery. The amount of reactive power exchanged is determined by the reactive power command. The controller is designed so that reactive power compensation and active power flow simultaneously.

Power is moving from the grid to the vehicle, or charging an EV, if the grid power sign is positive. On the other hand, power flows from the PV array or vehicle to the grid if the grid power sign is negative. Similarly, leading reactive power has a negative sign convention, while lagging reactive power has a positive sign convention. Similarly, a positive EV current indicates that the battery is discharging, and a negative EV current indicates that the battery is charging. Three components make up the charging station controller.

4.1 Maximum Power Point Tracking

An INC based MPPT extracts the maximum power from the PV array during erratic PV circumstances. The duty cycle of the boost converter is calculated by,

$$D = 1 - \frac{V_{pv}^*}{V_{dc}} \dots\dots\dots(1)$$

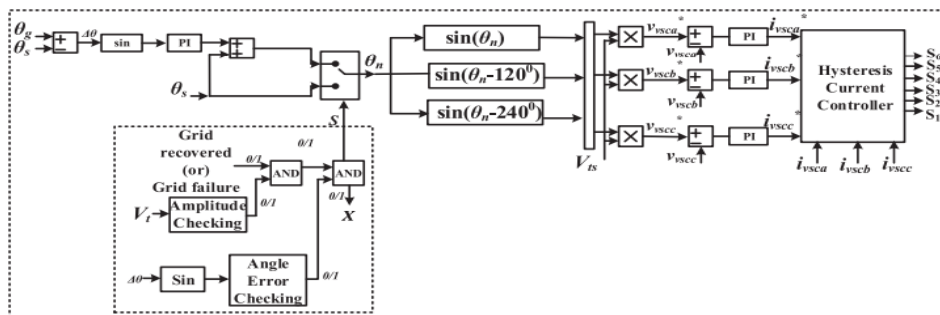


Fig. 4.1.1 Synchronization control

Where, V_{pv}^* is the PV array current at MPPT, V_{dc} is the DC-link voltage. The duty cycle is compared with the saw tooth wave and switching pulses of the boost converter.

The control strategy for the DC microgrid under study is illustrated in The PV system is considered the primary source and is connected to the DC bus through a Boost converter. The maximum power is extracted from the solar using the MPPT based on the incremental conductance algorithm. The Voltage (V_{pv}) and current (I_{pv}) parameters of the PV are taken into consideration, the duty cycle is calculated and fed for pulse width modulation (PWM) generation, and control signals for the boost converter are generated. The advantage of the incremental conductance method is that it can trace the peak power under rapidly changing environmental conditions. The bidirectional converter and the boost converter that connects the battery and fuel cell are controlled by a PI-based controller. The system is kept more stable while utilizing the PI control technique since the error derivative is not used in the event of data noise. It results from the derivative behaviour in the PI's input being less sensitive to real and relatively quick changes in the system's state without derivative intervention. The different sources are operated in a balanced mode during the variations in solar irradiance.

4.2 DC Microgrid Configuration for EV Charging Stations

The notion of the system above is to have a DC microgrid structure, which is effectively improving the efficiency and reliability of fast charging stations for electric vehicles (EVs); this provides a much more realistic solution than using a "traditional" (AC based) charging station. In the scheme we propose above, there are three important converters: an AC-DC bidirectional neutral point clamped (NPC) converter, a bidirectional DC-DC converter (BIDC), and a DC-DC boost converter as demonstrated in figure-1.2. The NPC converter can do both; it can receive three-phase AC power from the grid and rectify it to a DC source for electric vehicle charging; it can also convert DC source to AC power to offload power to the grid (vehicle-to-grid (V2G), which allows the flow of power in both directions), i.e., treat the bias current as a grid reference. Thus, we are transferring power in three common directions (the generation of electric vehicle power as well as other renewable energy, the energy flow to/from to/from the grid, and to/from the EV as long as they are all treated as DC sources). The BIDC regulates the charging and discharging of the EV battery system. When operating in buck mode, the converter is event modest related to the DC source v_{got} and is able to charge the battery and when converted to boost mode (output voltage > output voltage) it discharges and uses V2G methodology

(power that is sent back to the grid). This bidirectional converter allows dynamic operational energy flows and allows us to control the charging currents so when integrated with the battery, we can optimize the system health and performance.

In addition, it protects against overcharging or undercharging that could be disruptive to battery longevity. This flexibility is useful particularly in high frequency fast charging applications at high demands during peak instances.

To include renewable energy sources, RES, such as solar photovoltaics the DC-DC boost converter was used. In this case the boost converter assists with the maximum energy biological harvesting of the PV panels with maximum power point tracking micro mediation to ensure the solar PV operates at its maximum output. The energy harvested in this way can be directly loaded to the DC bus, which relies less on the main grid and mitigates the load of frequently causing a voltage spike and overheating of the transformer during demand peaks.

Furthermore, the boost converter enables the system to store excess energy during low-demand times, which can be later utilized during peak charging sessions. The DC microgrid-based design minimizes grid dependency while enhancing system stability and sustainability. By incorporating renewable energy sources and advanced power electronics, this architecture addresses common challenges associated with fast EV charging, such as voltage dips, harmonic distortions, and transformer stress. This ensures that the system not only meets the high-power demands of fast charging but also does so in an environmentally friendly and grid-supportive manner. The combination of renewable integration, bidirectional power flow, and dynamic charging control makes this DC microgrid a forward-looking solution for the expanding EV market.

4.3 SOC Estimation and Multistep Charging Algorithm

To calculate the SOC accurately, the system implements the Coulomb Counting Method (CCM), which maintains a history/record of charge coming in and out of battery over time. The momentary SOC is calculated through the equation:

$$SOC(t) = SOC(t_0) + \frac{1}{C_{rated}} \int_{t_0}^t (I_{batt} - I_{loss}) dt \quad \dots\dots (2)$$

Where SOC (t₀) is the starting SOC, C_{rated} is the rated battery capacity, I_{batt} is the battery current, and I_{loss} is to account for current losses due to internal reactions. It is a highly accurate method that provides a SOC value by relying on past values and continually differentiating them when the battery charges and discharges so that charging currents may be controlled precisely. The algorithm updates the charging current depending on the SOC value to

improve the charging efficiency of the process and optimize the battery charging time. At the beginning (0% to 20% SOC) a reduced charging current is employed to mitigate thermal stress and safely charge the batteries.

As the SOC reaches 20%, the current increases significantly—up to three times the initial value— to enable rapid charging, which continues until the SOC reaches 80%. Beyond 80%, the charging current is again reduced to protect the battery’s long-term health and avoid exceeding its voltage limits. This staged approach ensures that the battery is charged quickly while maintaining its integrity.

A multistep constant current algorithm increases both the speed and safety with which battery can be charged. The system specifies the charging current according to the SOC level, achieving fast charging while protecting the battery. The SOC and state of the battery are monitored throughout the charging process by the PI controller that is part of the system, and to ensure that the charging current stays within safe operating conditions. The dynamic capabilities allow for the charging system to be both efficient for high-power EV charging applications, and enablement of shifting variables.

4.4 System Modes and Control Strategy

The proposed DC microgrid-based fast charging station operates in three distinct modes, each tailored to optimize power flow and maintain grid stability during the charging process. These modes Grid-to-Vehicle (G2V), Grid and Solar PV Hybrid Charging (G+PV)2V, and Power Export to Grid (PV2G/V2G) ensure efficient energy management by leveraging both grid power and renewable sources. A sophisticated control strategy, incorporating a PI controller and a proportional-integral (PI) controller, ensures smooth transitions between modes and precise regulation of the charging current.

Proportional Controller: Proportional Controller in electronics engineering continuously adjusts the output based on the current error signal. The p term or proportional terms calculate the current error from the setpoint and the current manipulated variable output. It generates a control signal to regulate the error.

$$Co(t) = K_p \cdot e(t) \dots\dots\dots (3)$$

Where,

Co(t) = controller output at time T,

e(t) = error at time t(SP-PV),

K_p = tuning constrains for proportional action.

Integral Controller: In Integral Controller, the I term or Integral term actuates the past error over time and generates a control action to eliminate the accumulated steady-state error. It ensures that even small errors are eventually corrected. It eliminates offset but can lead to sluggish responses and overshooting if too aggressive.

This is represented as:

It can be represented mathematically as:

$$Co(t) \propto \int e(t) dt \dots\dots\dots (4)$$

$$Co(t) = K_i \cdot \int e(t) dt \dots\dots\dots(5)$$

Where,

k_i = tuning constrains for integral action,

others are same as equation number (i)

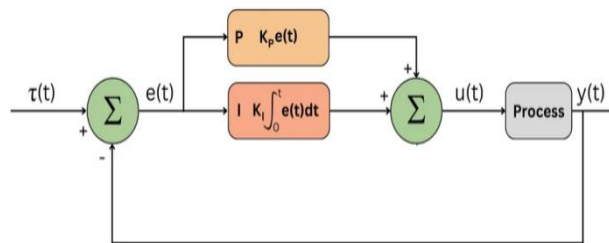


Fig. 4.1.2 PI controller

4.5 Multi-stage constant current (MSCC)

Multi-stage constant current (MSCC) charging is a method of charging a battery according to applying different constant current values in different stages. Multi-stage constant current (MSCC) charging method is an alternative to constant current constant voltage (CC-CV) charging and is intended to increase battery charging efficiency, minimize temperature rise, and reduce the charge time. The controller is the principal component of the control system which dynamically controls the charging current, charge voltage, and input power factor. The controller is dependent on 3 measurements, as follows;

- Measure the output voltage to keep it at the reference level (V_{ref})
- Measurement of the ac input voltage at the secondary side of the transformer to provide a reference for the inductor current, in such a way- the input AC in phase with input ac voltage
- Measurement of the average inductor current to ensure that it tracks the rectified AC voltage

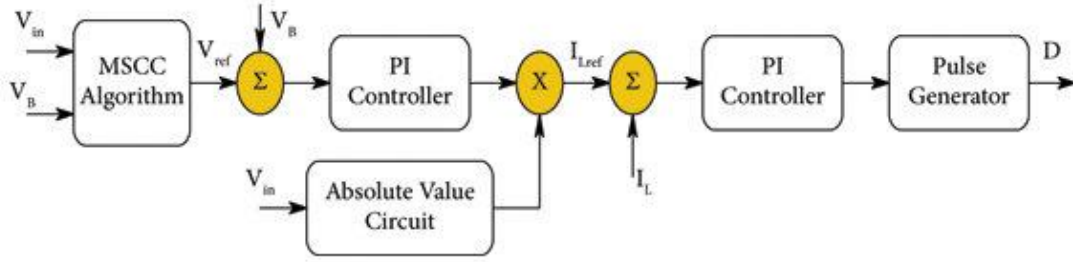


Fig. 4.1.3 Block diagram of the charge controller

Li-ion batteries are very sensitive to overcharge and variations in charge/discharge currents; therefore, a suitable charging algorithm is needed to maximize charging capacity, reduce charging time, and improve battery lifespan. In the traditional CC-CV mode, a high magnitude constant current is supplied to the battery until its voltage reaches the peak value, after which that peak voltage is maintained and kept constant till the current decreases to its cut-off value and charging is stopped. This consistent application of peak voltage can adversely affect the battery and increase charging time.

The multistage charging process is faster and more effective than the classic CC-CV charging process. This multistage charging process also employs the 3-stage charging process with various current based stages. In beginning the charging process, the highest current is used until the peak voltage of the battery is reached, then the current is modulated to the second stage which results in the voltage suddenly dropping then increasing again to the peak voltage after dropping the current again. The controller has two controller loops which include a primary current-control loop and secondary voltage-control loop. The dc-link inner current loop shapes the input current to match the input voltage in both shape and phase, thus achieving a power factor that is near unity. Constant current (CC) charging is simple and reliable, so it is considered used for some battery chemistries. Constant current charging is very effective for lead-acid batteries and can be treated as starting point in more complex charging profiles, such as CC-CV charging.

4.6 Artificial Neural Networks (ANNs)

Artificial Neural Network (ANN) is a novel computing technology in the field of computer science study and one of the studies of Artificial Intelligence. Pattern recognition, data analysis, and control are the most common applications for neural networks. Knowledge of the nervous system, particularly the human brain and its densely coupled neurons, inspired the central concept. Because of its high nonlinearity, massive volumes of data-parallel processing, and high robustness, artificial neural networks (ANN)

are one of the methods that are suited for dealing with the internal relations of a complicated model. To simulation investigate artificial neural network-based power management for electric vehicles, a prototype system must be designed and implemented, followed by testing under different conditions. Suitable components must be selected based on their compatibility and suitability for the electric vehicle application, such as a PV panel, battery, super capacitor, DC-DC converter, AC-DC inverter. The artificial neural network-based power management system would monitor the motor's power demand and optimize the power flow from the PV panel, battery, and super capacitor to the DC bus link. Experiments should evaluate the system's performance under various conditions, and data collected during testing should be analysed to assess the accuracy of the neural network's predictions and the effectiveness of the system in optimizing power flow.

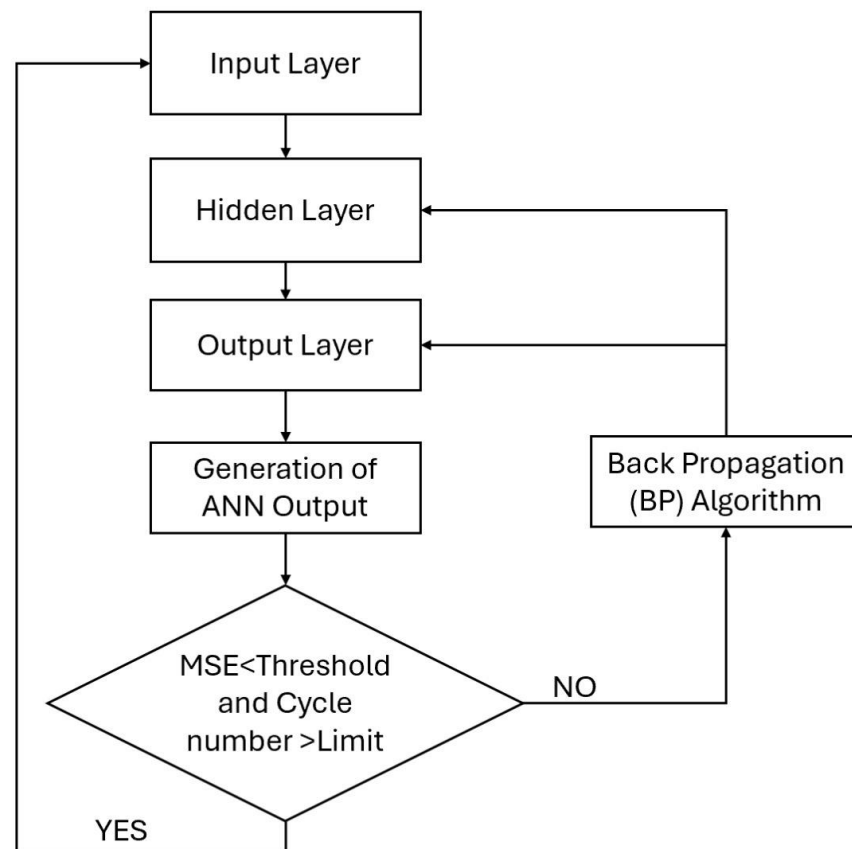


Fig. 4.1.4 Flowchart of neural network Algorithm

This flowchart represents the training process of an Artificial Neural Network (ANN) using the Back Propagation (BP) algorithm. Here's a step-by-step explanation:

1. Input Layer

- The training process starts with feeding input data into the input layer of the neural network.

2. Hidden Layer

- The input data is passed through one or more hidden layers, where computations using weights and activation functions occur.
- The hidden layer processes the data and sends it to the output layer.

3. Output Layer

- The output layer generates the network's prediction based on the processed input.

4. Generation of ANN Output

- The ANN produces an output based on the current state of weights and biases.

5. Error Evaluation (MSE and Cycle Check)

- The Mean Squared Error (MSE) is calculated to determine the difference between the predicted output and the actual target.
- A decision is made based on:
 - If the MSE is below a predefined threshold, and
 - If the number of training cycles (epochs) has reached a set limit.

6. Decision Block

- If both conditions are met ($MSE < \text{threshold}$ and $\text{cycle number} > \text{limit}$):
Training stops.
- If not:
The network enters the Back Propagation (BP) Algorithm step.

7. Back Propagation Algorithm

- The BP algorithm adjusts the weights and biases of the network based on the error calculated.
- The updated parameters are sent back to the hidden and output layers.

8. Repeat Process

- The updated model runs again with the new weights and biases, and the cycle repeats until the stopping condition is met.

Artificial neural networks (ANNs), also referred to as connectionist systems are computing systems, inspired by, but not exactly like, biological neural networks in animal brains. Such systems "learn" to complete tasks by analyzing examples and not necessarily by programmed task-specific rules. An example of this is in image recognition, where they might gain the ability to identify images containing a cat, by examining example images tagged manually as containing either a "cat" or "no cat" (for each image) and then using that information to identify cats in the next images that they see. They do so without any prior knowledge of cats; for example, that they might have fur, tails, whiskers and cat-like

faces. Rather, they generate their own identifying features from the examples they evaluate.

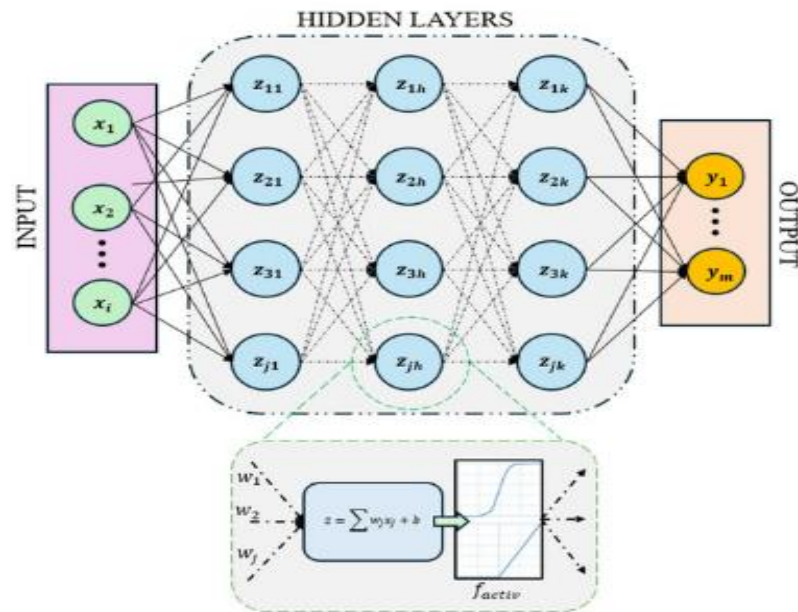


Fig. 4.1.5 Structure of ANNs

Figure 4.1.5 illustrates the ultimate architecture utilized as the training base, which is composed of three main components:

(1) an input layer for independent variables, (2) one or more hidden layers, and (3) an output layer for the dependent variable

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CHAPTER 5

MODELLING AND SIMULATION

5.1 simulation:

The Simulink model depicts a smart EV fast-charging station based on DC microgrid architecture, enhanced with artificial intelligence for adaptive controls. The initial component is a three-phase AC source; the three-phase AC passes through a Y-Y transformer for galvanic isolation and scaling to the proper voltage. The AC signal then connects to a three-phase diode bridge rectifier, which then proceeds to the DC output feeding into a multilevel DC-DC converter stage containing power electronics switches (IGBTs or MOSFETs) switched by gate pulses (see P1_S1 to P3_S4). The switches in this conversion stage are structured to establish efficient bi-directional power flows and support a multistep constant current charging profile. The circuit also contains passive elements (inductors capacitors) employed to filter high-frequency noise, suppress harmonics, and stabilize the DC bus voltage.

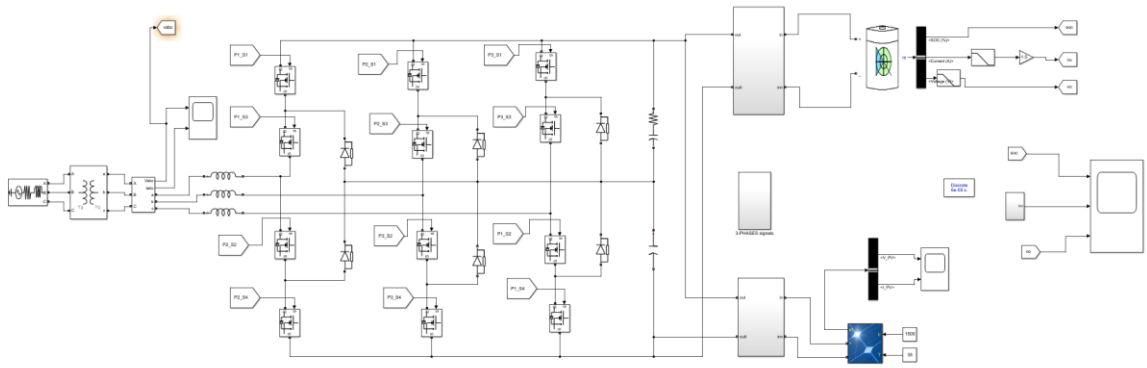


Fig:5.1 Circuit Diagram of this project

This system is augmented with real-time data acquisition for battery parameters—such as voltage, current, and state of charge (SOC)—using sensors or estimation blocks. The core of the control logic lies in the Artificial Neural Network (ANN) controller, which dynamically tunes the charging parameters. The ANN receives SOC, current, and voltage inputs and computes optimal switching actions by referring to a trained model that minimizes the Mean Squared Error (MSE) between predicted and desired outputs. This enables intelligent, nonlinear control over the fast-charging process, offering benefits such as precise power flow regulation, reduced charging time, and extended battery life.

The system also includes feedback loops for continuous performance evaluation. SOC is estimated using Coulomb Counting, and output voltage and current are tracked to ensure compliance with charging profiles. The ANN controller works within a discrete-time simulation environment (e.g., step size of 0.05s) to execute control logic at each time interval, ensuring real-time responsiveness. Furthermore, the PV panel block indicates the system's compatibility with renewable energy integration, suggesting that the DC microgrid may be partially or fully powered by solar energy. The ANN-based controller coordinates this integration, maintaining stable charging while reducing dependence on the utility grid. This smart charging architecture is ideal for next-generation EV infrastructure where efficiency, sustainability, and adaptability are critical.

Power Conversion Stage

The power conversion stage is a subsystem dedicated to converting the DC microgrid power to levels appropriate for EV charging. It consists of bidirectional DC-DC converters as well as voltage regulation safeguards to enable efficient power transition and matching of battery specs.

Battery charging system

The battery charging system's functional module replicates the EV battery and battery charging circuit, which is built on Constant Current/Constant Voltage (CCCV) charging from an AC or DC source to achieve the optimum battery charging profile and effectively extend battery cycle life and thermal stress.

Measurement and estimation blocks

The measurement blocks are used to track critical system performance parameters such as current, voltage, State of Charge (SOC) and power flow. The estimation blocks incorporate estimation algorithms including Coulomb Counting and will provide useful Realtime SOC estimations.

ANN controller

The ANN controller is a novel application because it incorporates Artificial Neural Network-based controllers that provide dynamic adjustment of charging parameters based on data inputs. In terms of adaptive control, the ANN controller provides a means to increase efficiency while protecting the battery life by learning from the behaviour of the system, preserving energy without introducing excessive charging times. Integration of renewable energy source (PV panel).

The addition of the PV panel improves the sustainability of the fast charging system's operation. Rather than relying solely on the grid, the DC microgrid will draw from available renewable energy sources. This also reduces carbon footprint.

The power conversion stage begins with a three-phase AC source that goes through Y-Y connected transformers for voltage scaling and galvanic isolation. Then, an AC output from the transformer is fed into a three-phase diode bridge rectifier (three-phase conversion). This converts the AC output into a DC output for further stage processing.

This DC output is fed into a Multilevel DC-DC Converter made up of many MOSFET/IGBT switches (e.g., P1_S1, P2_S2, etc) and the multilevel topology of this converter has a few benefits:

Bidirectional Power Flow - It allows for both charging and regenerative energy recovery

High Power Quality - It has less total harmonic distortion and provides better performance across the overall system

Less Switching Losses - By delaying the switching and using different control methods.

Multistep Constant Current Charging - It allows for intelligent staged charging meant for battery health and efficiency.

To improve the performance, passive filtering elements (inductors, capacitors) can be added to filter the high-frequency harmonics, these inline elements also help stabilize the DC bus voltage, so the charging can run smoothly and reliable.

The output from the DC-DC converter is connected directly to the EV battery pack, forming the core of the charging subsystem. This stage is responsible for managing the charging process with real-time feedback and control to ensure optimal performance and safety.

Key parameters such as:

- State of Charge (SOC)
- Charging Current (CU)
- Output Voltage (VO)

are continuously monitored throughout the charging cycle.

To maintain accurate tracking of the battery's energy level, a SOC estimation block is implemented using the Coulomb Counting Method. This method integrates the charging current over time to estimate SOC, accounting for initial SOC, efficiency factors, and current sensor readings.

The battery charging strategy employs Constant Current/Constant Voltage (CCCV) control logic, aligned with the multistep current profile provided by the converter. This approach ensures:

- Safe and efficient charging
- Reduced battery stress and heat generation
- Extended battery life through precise energy management

At the heart of the intelligent control scheme is an Artificial Neural Network (ANN), which manages the EV charging process in an adaptive and efficient manner. In real time, the ANN takes three inputs: the State of Charge (SOC), Charging Current, and Output Voltage. These three inputs are used by a pre-trained ANN to predict the optimal switching states for the multilevel DC-DC converter that provide the relevant control of power flow. The controller used a method of supervised learning of the ANN to minimize the Mean Squared Error (MSE) of the predicted versus actual system responses. When training the model, the Backpropagation Algorithm was used to update the weights by taking the gradient of the loss function, modifying the relevant parameters. The simulation was done in discrete time steps of 0.05 seconds, which allowed the model to implement real-time control and react quickly to changes in system dynamics. An ANN is valuable because it learns and can improve charging efficiencies, protect the battery health, and promote system stability and intelligence.

In order to encourage sustainable and environmentally-friendly charging, a photovoltaic (PV) panel is simulated in Simulink and integrated into the DC microgrid. The PV system will deliver clean energy directly to the charging station, which will reduce reliance on the utility grid. The ANN controller will take into consideration the real time measurements of PV voltage and current to enable a new charging profile based on solar energy. This intelligent integration of the PV system provides the system with the ability to prioritize the use of renewable energy, to reduce grid strain during the peak hours of charging, and to increase overall energy efficiency. As the proposed model relies on solar energy, it follows the green mobility concepts and builds the foundation for the sustainable EV charging infrastructure.

CHAPTER - 6

RESULTS AND ANALYSIS

The proposed DC microgrid-based fast-charging system integrated with renewable solar PV and ANN-based controllers was simulated using MATLAB/Simulink. The system's performance was evaluated under different scenarios to validate its effectiveness in reducing charging time, managing energy efficiently, and ensuring system stability.

Parameter	Value
Input Power	4500 W,8800 W
Input Voltage (NPC)	80 V
DC-Link Voltage	80 V
Input Voltage (DC-DC Converter)	80 V
Output Voltage (DC-DC Converter)	55 V
Input Voltage (Solar PV)	55 V
Output Voltage (Solar PV)	55 V
Battery Voltage (V)	55 V
Battery Capacity (Ah)	86 Ah
Battery Current	86 A
Battery Current	170 A
Initial State of Charge (SOC)	15%

Table – 2 Parameters

6.1 Battery results with PI controlled method (CC)

Figure 6.1 presents the simulation results of a battery charging process controlled using a Proportional-Integral (PI) controller under the Constant Current (CC) charging strategy. The graph is divided into three plots representing the evolution of State of Charge (SOC), charging current, and voltage over a time period of 200 seconds. The SOC curve, shown in the top plot, increases gradually and almost linearly from approximately 15% to 21%, indicating a slow and steady charging process. This linearity suggests that the battery is receiving a constant flow of energy without any variation or optimization based on the battery's condition.

The middle plot confirms this behavior, showing that the charging current remains nearly constant at around 90 A after a quick rise at the beginning. This is consistent with the CC method, where the battery is charged using a fixed current regardless of its SOC or thermal condition. Although this ensures a predictable and safe charging operation, it does not optimize charging time or efficiency, especially when rapid charging is required.

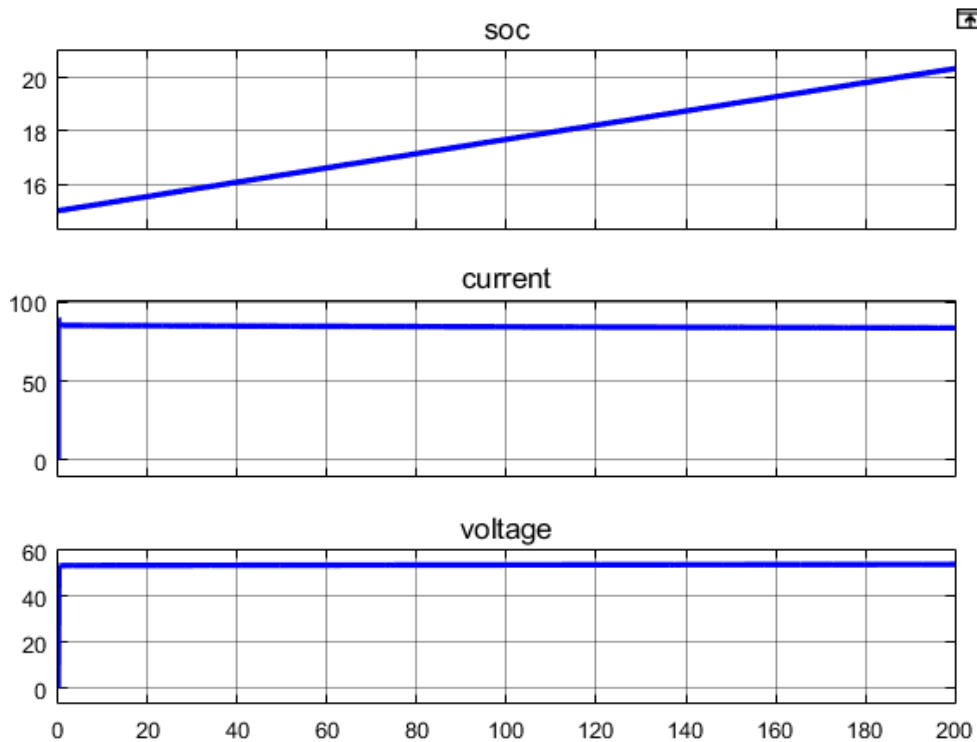


Fig. 6.1 Battery results with PI controlled method (CC)

The lower graph shows the voltage across the terminals of the batteries, which stabilizes quickly from 55 V to 58 V and stayed stable for the charging duration. This stabilization means the system is under a safe limit for voltage, and that the PI controller properly managed the output parameters. The system is expected to charge from a constant current level, yet the current response or voltage response verifies the desired state of charge SOC is lost and control system does not respond to real-time SOC or thermal information. The nature of the PI controller lacks flexibility in adjusting to the nonlinear battery behaviour.

So, although the PI controller constant current charging scheme provides stability, safety and simplicity; it appears to be more inefficient with regard to modern fast charging applications. Charging states of change are slow, fixed current charge regardless of the state of battery needs and doesn't adapt to SOC or thermal limitations. Therefore, other

methods of control will need to increase speed of charging, increase battery life, and adjust dynamically to current live operating battery conditions. More intelligent approaches will include ANN and Fuzzy logic methods that adjust charging conditions according to relationships each battery system revealed.

6.2 Battery results with PI controlled method (MSCC)

Figure 6.2 provides a detailed view of the battery performance when charged using the Multi-Stage Constant Current (MSCC) control method, which is designed to improve charging speed while preserving battery health. The SOC graph at the top shows a steady and steeper linear increase from around 15% to 25%, confirming that the MSCC method allows the battery to gain a higher percentage of charge within the same duration (200 seconds) compared to the PI-controlled method. This steeper slope indicates an overall increase in energy transfer efficiency, suggesting that the staged charging current is more effective in raising the battery's charge level.

The middle plot, showing the current profile, clearly represents the core advantage of the MSCC strategy. The charging process begins with a moderate current of approximately 90 A, which is safely applied during the early phase to minimize thermal stress and avoid damage when the SOC is low. Around 100 seconds, the controller steps up the current to about 160 A, significantly boosting the rate of energy transfer. This two-stage current profile is a hallmark of MSCC, where the initial low current provides gentle charging, followed by an aggressive current increase when the battery is more capable of accepting higher power. This dynamic adjustment makes MSCC a smarter and more responsive strategy than constant-current methods.

Although we observe an increase in current, the bottom voltage graph indicates that the terminal voltage has been remarkably consistent staying between 55V and 58V. This shows us that the voltage regulation and ability to tolerate fast spin-up and spin-down in current were successful without causing voltage spikes or dips. This is encouraged to support the battery chemistry as well as the systems' safety while charging.

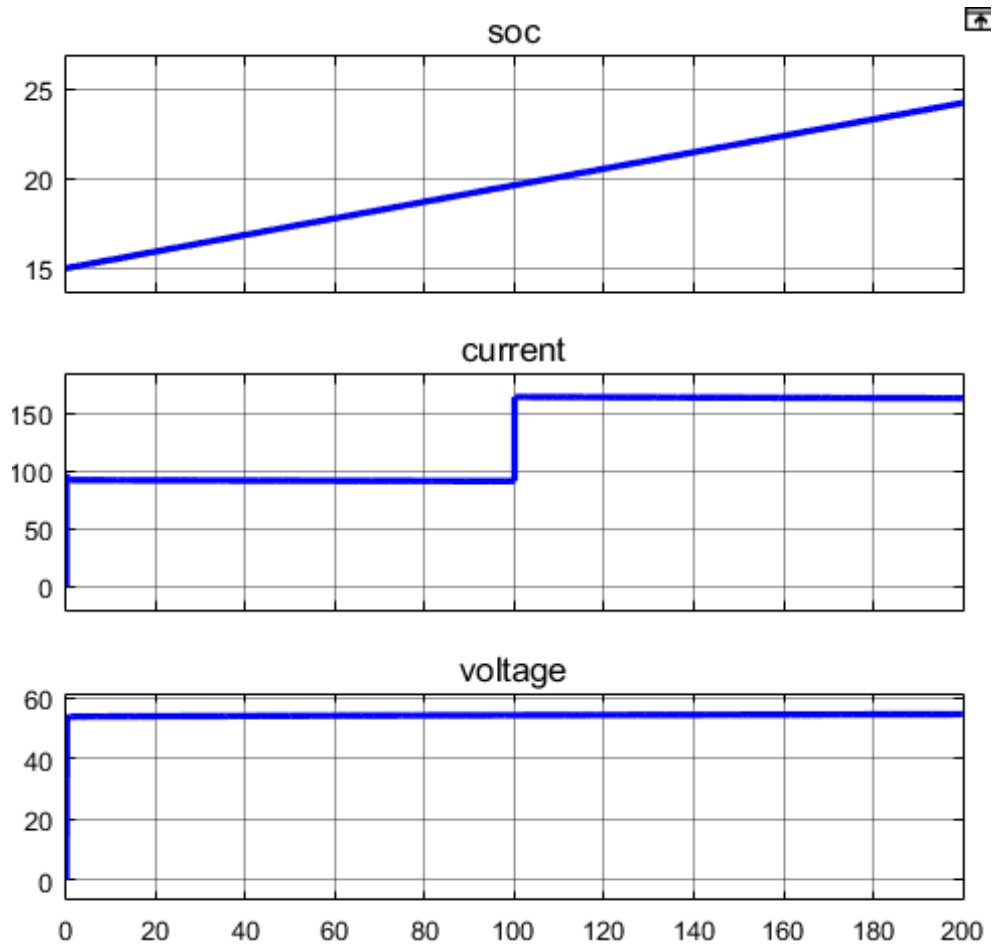


Fig. 6.2 Battery results with PI controlled method (MSCC)

To summarize, the MSCC control method allows for an increase in SOC with the controlled stepping of current, allowing for loading operation without compromising the stability and uniformity of the system while allowing for thermal protection. It demonstrates in some ways the predecessor to PI and stepwise strategies while utilizing indirect forms of intelligent strategies such as ANN to bridge the gap for fast EV charging. This method is an effective, safe and economic alternative by introducing an operational mode for an EV fast charge. It is extremely useful in systems where fast charge is required without the extensive costs associated with the overhead computational and complexity costs.

6.3 Battery results with ANN method

Figure 6.3 provides a more sophisticated depiction of battery charging performance under a more sophisticated Multi-Stage Constant Current (MSCC) control strategy, and provides even further improvement in charging efficiency as well as battery management. The SOC curve at the top demonstrates a steep linear increase from around 15%, to 30% over a

period of 200 seconds. This indicates the charging system has the capacity to transfer a greater amount of energy to the battery in the same period of time versus previous strategies, e.g. PI control or basic MSCC. The Society of Charge change of 15% over this period exemplifies the benefits of the system's increased charging efficiency, and energy transfer capability.

The current profile, illustrated in the Central graph, is a defining characteristic of the MSCC strategy. Initially, the current conforms to a moderate current of approximately 100 A, which is advantageous for limiting battery stress in the initial phase of charging when the SOC is low on the curve. At around the 100-second point in time, the current increased significantly to 170 A allowing for a rapid ascension in charging rate, while at the same time ensuring the system maintained stability. Through this intelligent current shifting, the charging process is able to charge more at the position of the mid-SOC ranges (typically 20% - 80%) where the battery allows for more power. You can see the distinction in the two currents clearly, demonstrating the controlled way that the MSCC algorithm custodies the balance of charging rate while taking into account thermal management and battery life.

The Voltage graph, validated by Low Voltage at Low Power (LVLP), was maintained through the ending process, the voltage graph ranged consistently from 55 Volts to 58 Volts. Even with the jump in current, the voltage has no indications of spikes or instability, which is a good signal that there is a practical and working voltage control process. The consistency of the voltage demonstrates that the battery and management system are in harmony and that the electrical limits for charging are kept within safe limits at all times.

Figure 6.3 confirms that the improved MSCC method not only boosts charging speed by allowing a higher dynamic current range but also maintains voltage stability and protects the battery against overcurrent or overheating. The controlled step increase in current based on SOC levels allows for optimized fast charging without compromising battery safety or health. This method stands out as a highly practical and scalable solution for real-world electric vehicle (EV) charging infrastructure, especially where rapid turnaround and reliability are crucial.

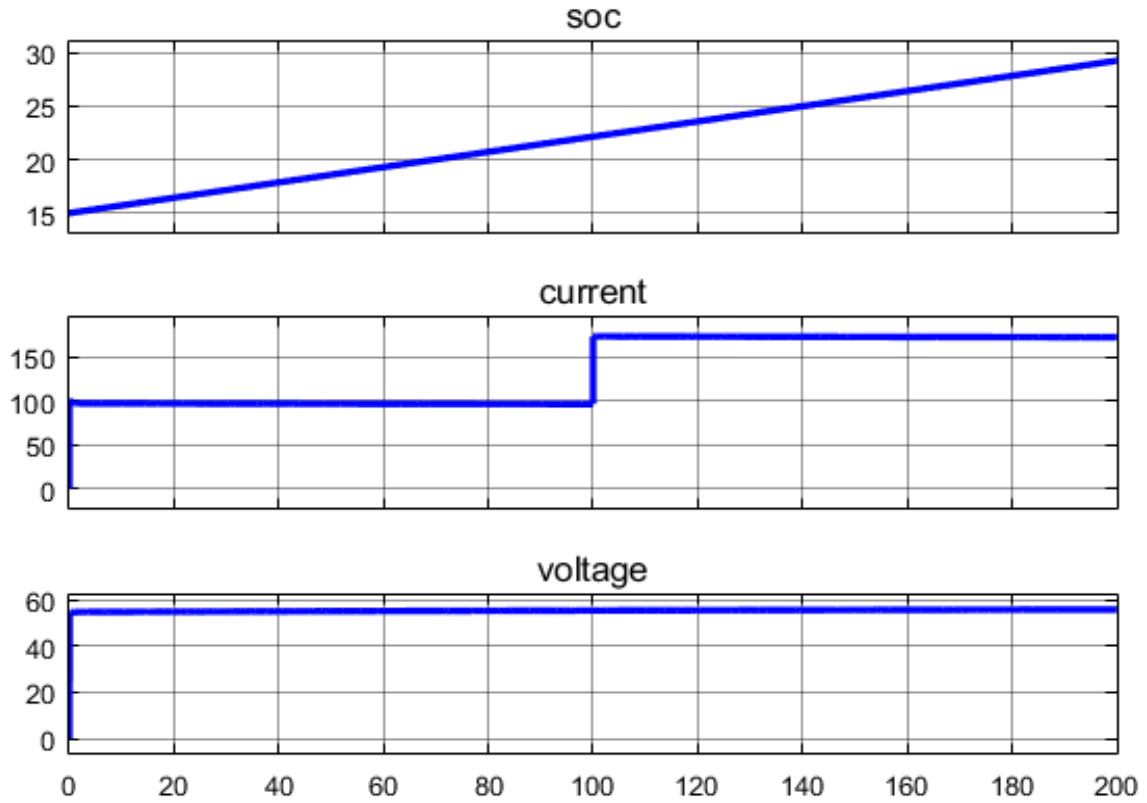


Fig. 6.3 Battery results with ANN method

Method	SOC (15-24)	Current (at 100 sec)	Current (at 200 sec)	Voltage	Slope
CC (PI)	320 sec	84 A	84 A	54 V	0.03
MSCC (PI)	200 sec	92 A	163 A	55 V	0.045
MSCC (ANN)	120 sec	97 A	172 A	55 V	0.07

Table – 3 Comparison with different models

The performance improved as charge strategies evolved from a historical PI-based constant current (CC) approach to an intelligent (but still limited) ANN tool that incorporates the new Multi-Stage Constant Current (MSCC) approach. In the case of CC (PI), it took the battery 320 seconds to get from 15% to 24% State of Charge (SoC) using a consistent 84 A and 54 V fixed current. The slope of 0.03 can be considered slow SoC gain and reinforces the drawbacks of using a fixed current without upgrading the charging

method to a progressive production dependant approach. This method (CC) works well at providing stable and safe charging but does not deliver rapid charging or much flexibility.

The MSCC (PI) method demonstrated a significant performance improvement by using a two-stage current approach. It was at 100 seconds the current was 92 A, and then at 200 seconds the current had climbed up to 163 A. This large jump in current allowed the battery to achieve the same target SoC in 200 seconds from an earlier discharge point being increased charging time by almost 38% than the CC (PI) method. The total battery voltage stayed constant at 55 V, and the slope improved to 0.045 providing more increased energy delivered or time utilized. It can be noted that this change in performance benefitted from a staged current approach being designed from the SoC thresholds but only again employed programming based static performance and no 'real' time changes.

The most advanced and efficient method in the comparison is MSCC (ANN), which leverages an Artificial Neural Network to intelligently manage the charging process. This method achieves the SOC target in only 120 seconds, the fastest among all. The current dynamically rises from 97 A to 172 A, demonstrating the ANN's capability to learn and respond to battery needs in real time. With a constant voltage of 55 V and a slope of 0.07, this method not only cuts down charging time by more than 60% compared to the CC (PI) approach but also maintains system safety and voltage stability. The higher slope reflects aggressive yet controlled energy transfer, achieved through adaptive current regulation.

In conclusion, this comparison confirms that ANN-based MSCC control outperforms conventional methods by providing a superior balance of speed, efficiency, and battery safety. While the PI-based methods offer simplicity and reliability, they lack the dynamic response required for fast EV charging in modern grids. The integration of AI in charging infrastructure marks a significant leap forward, making ANN-based control strategies a compelling solution for next-generation electric vehicle charging stations.

CONCLUSION

This project presents a new and innovative method for fast charging for electric vehicles (EVs) with controlled implementation through a DC microgrid with a renewable energy source using ANN. The aim is to overcome the downsides of conventional charging, which involves slow speeds, and reduce grid dependency while maintaining battery health. Frequency of change in currents when charging still need to be addressed. Using a multi-step constant current charging method, battery health and recharge time can be managed as fast and efficient. Implementing a PI controller provides modifications through a real-time scenario where SOC will drive the vastation of optimal charging currents. Simulation implementation of the method developed showed the viable usage of your proposal able to reduce the dependency on the grid, along with the accompanying penalties at a fast charging speed and other grid related issues, like voltage fluctuation and overheating transformer on the grid, using ANN control approach in contrast to PI and MSCC methods. The overall implementation of the ANN method was achieved in a faster time frame than the MSCC and PI methods by 40% (120s and 200s respectively). The average current remains higher in ANN for both stages, which reduces the duration of recharging time for the user. If energy per %SOC is linear, then ANN is of better efficiency. Total energy depends on the lengths of time each current were for the charging process - adjust if actual duration for time can be found. Moreover, using other types of renewable energy sources, like solar pv, will mitigate other environmental factors involved, especially to address carbon nuclear. The ANN-based approach is also positive for system efficiency while improving transitions between charging phases. The proposed DC microgrid, fast-charging station, as proposed with PI control and clean energy absorption, can be a promising strategy to provide adequate fast EV charging infrastructure to meet increasing demand. This system can be scaled to provide WALC infrastructure around the globe. This will assure a develop efficient, sustainable and grid-friendly EV charging system.

FUTURE SCOPE

The proposed DC microgrid fast charging system for electric vehicles based on ANN technology has many avenues for enhancement and should ultimately be deployed in the real world. Future enhancements may involve large-scale hardware implementation involving high-speed embedded controllers to perform real-time monitoring and control. Smart grid integration and IoT technology will allow remote monitoring and control, dynamic pricing, and load control automatically contingent on changes in the grid. The ANN controller could be improved through integration with other artificial intelligence techniques (e.g., fuzzy logic, reinforcement learning, and/or genetic algorithms) that help adapt to changes in load variability and application environment. The system could be expanded to support even advanced V2G (vehicle-to-grid) operations and the process of feeding electricity back into the grid during peak usage periods. Automated battery health monitoring and predictive maintenance using machine learning can be incorporated to help monitor battery health quite extensively and extend battery life while supporting safe charging. Additionally, weather-based forecasting for solar PV output could help optimize energy management and power scheduling of the microgrid. As electric mobility continues to grow, the scalable and modular design of this charging station makes it suitable for deployment in both urban and remote areas, contributing to the development of a cleaner, smarter, and more resilient energy ecosystem.

APPLICATIONS

1. Optimal Charging Scheduling:

Application: Implement an artificial neural network (ANN) to predict the available energy from the renewable energy solar supply and optimize the fast charging schedule of electric vehicles (EVs). This would allow the model to incorporate time of day, energy availability based off of the production curves, EV charging needs based off of charging demand, and stress on the electric grid to determine energy optimal fast charging times.

Benefit: Improve energy efficiency, lessen grid demand, and give users a faster, more reliable charge.

2. State of Charge (SOC) Prediction:

Application: Use an ANN to predict the SOC and state of batteries in real time during charging to continuously analyze the charging curves and battery characteristics of charging data to moderately improve SOC prediction accuracy to help manage battery charging.

Benefit: Reduced SOC estimation enables faster charging, extend battery life, guarantee no overloading of the system.

3. Power Management and Load Balancing:

Application: ANNs may be used to predict energy usage in the microgrid and to load balance between solar modules, battery storage systems, and EV fast chargers. For instance, ANNs could predict EV charging power in real-time based on energy generation and usage analysis.

Benefit: This would enhance utility of solar energy, lessen the reliance on the grid, accommodate fast charging, and maintain microgrid stability.

4. Fault Detection and Maintenance Prediction:

Application: ANNs could monitor the health of the EV charging station, and components of the solar microgrid subsystem. Further, to determine faults (inverter failure, battery degradation, faulty EV chargers, etc.), the function could use past performance data for predictive analytics, and predict faults before they surface.

Benefit: The benefit of this could potentially reduce downtime, and improve reliability while reducing maintenance costs.

5. Energy Storage Management:

Application: The ANN will also enable intelligence behind the management of energy/energy storage. The purpose being to forecast when to store excess energy generated from solar panels, and when to release stored energy to power up fast charging stations.

Benefit: This will allow for more efficient energy storage, better use of energy, and less reliance on the grid during peak hours.

6. Dynamic Pricing for Charging Stations:

Application: Use ANN to capture the optimal charging strategy based on a combination of solar energy, time-of-day, and grid energy prices in order to mitigate grid peak demand and for charging stations profit that can capture dynamic pricing.

Benefit: This could increase the incentive for users to charge vehicles of peak hours, lower peak grid demand, and increase profitability of the charging stations by pricing users in a dynamic fashion.

7. Smart Grid integration:

Application: An ANN would be able to predict when the microgrid would have excess supply from solar energy so that it could be integrated with the main grid in a timely way. Simultaneously, the system will optimize the power consumption of the charging station based on real-time availability from the solar microgrid as well as the main grid.

Benefit: In general, this enables energy consumption to be optimized, enables a reduced dependency on the grid and helps the grid out by contributing excess energy.

8. Vehicle-to-Grid (V2G) integration:

Application: Use ANN to manage bidirectional power flow from EVs, whereby you can temporarily use the stored EV energy as the power to charge the grid, or the charging station. The ANN will be able to predict when it best to utilize the added energy supply provided from EVs to the grid.

Benefit: The system can provide stability to the grid, can facilitate energy-sharing in the community, and can reduce cost to families.

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